Open Source Software and GitHub

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Introduction

Open source software is software whose source code, the human-readable instructions written by programmers, is made freely available to the public, so that anyone can view, use, modify and share it under a licence. This is in contrast to proprietary or closed-source software, such as Microsoft Windows or Adobe Photoshop, whose code is kept secret and whose use is controlled by the owner. Open source is built on collaboration: thousands of developers around the world contribute improvements, find and fix bugs, and add features. Today, open source powers most of the internet and the modern technology world: most web servers, all of the world's top supercomputers, the majority of smartphones and the cloud run on open source software, and nearly all modern software contains open source components.

What Is Open Source Software?

Open source software is software distributed with a licence that gives users the freedom to run the program for any purpose, study how it works through the source code, change it, and redistribute copies, including modified copies. Famous examples include the Linux operating system, the Android Open Source Project, the Mozilla Firefox browser, the Python and Java programming languages, the Apache web server, WordPress, the VLC media player, LibreOffice, Visual Studio Code, Blender, and the artificial intelligence frameworks TensorFlow, PyTorch and Hugging Face Transformers. Common open source licences include MIT, which is very permissive, Apache 2.0, GPL (GNU General Public License), which requires that derivative works are also open ("copyleft"), and BSD. Free software and open source are closely related: the Free Software Foundation, started in 1985 by Richard Stallman, who launched the GNU project in 1983, champions "free as in freedom", and the term "open source" was adopted in 1998 and promoted by the Open Source Initiative. In 1991, Linus Torvalds, a Finnish student, released the first version of the Linux kernel, which became the heart of the open source movement. Open source does not necessarily mean zero cost, though much of it is free.

Benefits of Open Source

Open source offers low cost, transparency, security through many eyes reviewing the code, flexibility to customise, freedom from vendor lock-in, community support, fast innovation, and an excellent way for students to learn by reading and contributing to real code. Contributing to open source builds a portfolio that employers value. Programmes such as Google Summer of Code and Hacktoberfest encourage students to contribute.

What Is Git and How Is It Different From GitHub?

Git is a distributed version control system, created by Linus Torvalds in 2005 to manage the development of the Linux kernel. It is a free, open source tool that you install on your computer, which tracks changes to files, especially source code, over time, so that you can see who changed what and when, go back to an earlier version, work on different features at the same time using branches, and merge changes from several people without overwriting each other's work. Each developer has a full copy of the project history on their own computer, so most work can be done offline. Common commands are git init (start a repository), git add, git commit (save a snapshot), git branch, git merge, git clone (copy a repository), git push (send commits to a remote repository) and git pull (get updates).

GitHub is a web-based platform, launched in 2008, that hosts Git repositories in the cloud and adds features for collaboration: pull requests, issue tracking, code review, project boards, wikis, GitHub Actions for automation, GitHub Pages for websites and GitHub Copilot for AI coding help. Microsoft bought GitHub in 2018 for about 7.5 billion US dollars. In short, Git is the tool (the version control software), and GitHub is a service (a website that stores and shares Git repositories and makes collaboration easy). Alternatives to GitHub include GitLab and Bitbucket, which also use Git. GitHub has over 100 million developers, and India has one of the largest and fastest-growing developer communities on the platform, with tens of millions of users.

What Is a Pull Request?

A pull request (PR) is a request to merge your changes from one branch into another, usually from your own copy (fork) or branch of a repository into the main project. It is a central feature of collaboration on GitHub. The workflow is as follows: you fork the repository (make your own copy) or create a branch, make your changes and commit them, push the branch to GitHub, and open a pull request, which describes what you changed and why. The project's maintainers and other contributors can review the code, leave comments, ask for changes, and run automated tests. Once approved, the maintainer merges the pull request, and your changes become part of the project. Pull requests allow quality control, discussion and learning. Good pull requests are small, focused, have a clear title and description, follow the project's guidelines (CONTRIBUTING file) and link to the related issue. An issue is a report of a bug or a request for a new feature.

Basic Workflow for Students

Create a GitHub account, install Git, create a repository (a project folder with a README file), add files, commit and push. To share files, such as notebooks, datasets or PDF documents, in a GitHub folder, upload them to the repository, and use the "raw" file link to access a file directly from code. A README.md describes the project, and a .gitignore file tells Git which files to ignore. Branch protection and licences should be added for serious projects.

Open Source in India and Tamil Nadu

The Government of India adopted a policy on open source software in 2015 to encourage its use in government. Projects such as Bhashini for Indian languages and parts of the India Stack use open source principles. Chennai has an active free software community, such as the Free Software Foundation Tamil Nadu (FSFTN) and groups that promote Linux and Tamil computing, and companies like Zoho contribute to open technologies. Tamil language tools such as Tamil Linux keyboards, fonts and Tamil text corpora are developed by the community. Many students in Chennai participate in hackathons and Google Summer of Code.

Challenges and Future

Challenges include funding and maintainers' burnout, security of the software supply chain, and licence compliance. Open source AI models, with open weights, are expanding access to artificial intelligence. The future of open source is closely tied to cloud, AI and collaborative development.

Quick Facts

- Open source software has publicly available source code that anyone can view, modify and share under a licence.
- Git is a version control tool created by Linus Torvalds in 2005; GitHub is a web platform that hosts Git repositories and enables collaboration.
- A pull request is a request to merge your changes into another branch or project, reviewed by maintainers.
- Linus Torvalds released Linux in 1991; Microsoft acquired GitHub in 2018.
- Common licences are MIT, Apache 2.0 and GPL.